

## Day 10 Of Communication System Class

### 8 Error CONTROL

8.1 Basic Terms Used In Error Control : Hamming weight, hamming distance, code vectors, constraint length, code rate, syndromes

#### 8.2 Error detection

##### 8.2.1 Parity Check

##### 8.2.2 Cyclic Codes

##### 8.2.3 Checksum

##### 8.2.4 CRC-n ( $n = 3$ to $32$ )

#### 8.3 Error correction codes:

##### 8.3.1 Linear block codes

##### 8.3.2 Hamming codes

##### 8.3.3 Convolutional code

## Introduction

- During transmission process, the transmitted signal passes through some noisy channel. Due to noise interference some errors are introduced in the received data. These errors can be detected and sometimes corrected using coding technique.

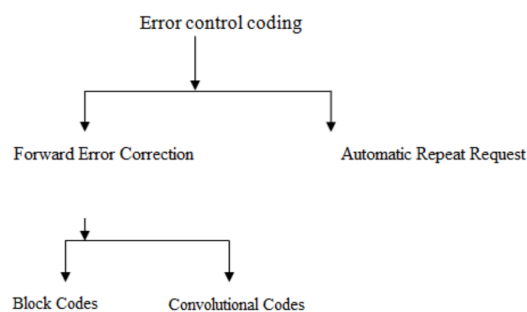


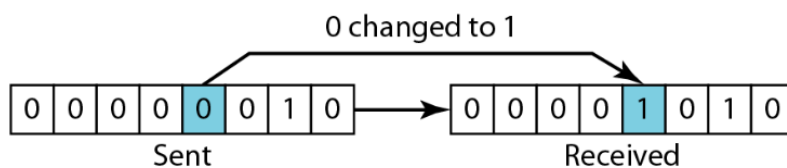
Figure: Types of Control Coding Techniques

**Data can be corrupted during transmission.**

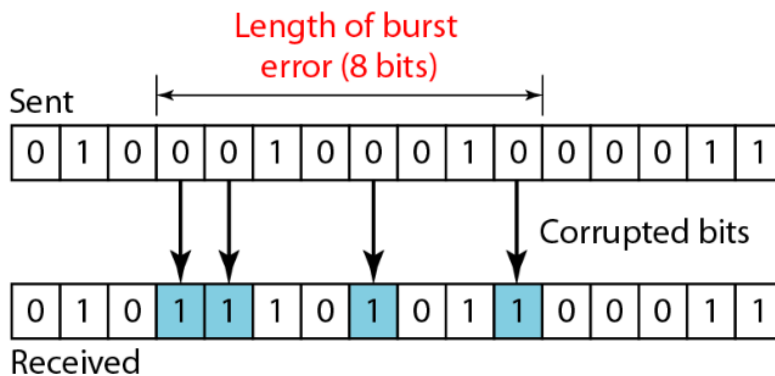
**Some applications require that errors be detected and corrected.**

Types Of Error :

**In a single-bit error, only 1 bit in the data unit has changed.**



**A burst error means that 2 or more bits in the data unit have changed.**



**To detect or correct errors, we need to send extra (redundant) bits with data.**

## 8.1 Basic Terms Used In Error Control

The Hamming distance between two words is the number of differences between corresponding bits.

The minimum Hamming distance is the smallest Hamming distance between all possible pairs in a set of words.

To guarantee the detection of up to  $s$  errors in all cases, the minimum Hamming distance in a block code must be  $d_{\min} = s + 1$ .

To guarantee correction of up to  $t$  errors in all cases, the minimum Hamming distance in a block code must be  $d_{\min} = 2t + 1$ .

**Table 10.4** *Hamming code C(7, 4)*

| <i>Datawords</i> | <i>Codewords</i> | <i>Datawords</i> | <i>Codewords</i> |
|------------------|------------------|------------------|------------------|
| 0000             | 0000000          | 1000             | 1000110          |
| 0001             | 0001101          | 1001             | 1001011          |
| 0010             | 0010111          | 1010             | 1010001          |
| 0011             | 0011010          | 1011             | 1011100          |
| 0100             | 0100011          | 1100             | 1100101          |
| 0101             | 0101110          | 1101             | 1101000          |
| 0110             | 0110100          | 1110             | 1110010          |
| 0111             | 0111001          | 1111             | 1111111          |

## 10-3 LINEAR BLOCK CODES

*Almost all block codes used today belong to a subset called **linear block codes**. A linear block code is a code in which the exclusive OR (addition modulo-2) of two valid codewords creates another valid codeword.*

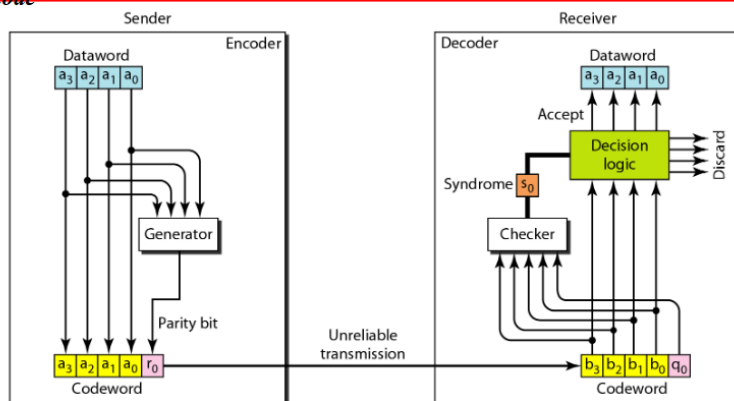
## 8.2 Error detection

### 8.2.1 Parity Check

**A simple parity-check code can detect an odd number of errors.**

**Figure 10.10** Encoder and decoder for simple parity-check

~~code~~



Q. Explain How 2D Parity Check Works ?[ Most Probable Question ]

**Figure 10.11** Two-dimensional parity-check code

|                 |   |   |   |   |   |   |   |   |              |
|-----------------|---|---|---|---|---|---|---|---|--------------|
| 1               | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | Row parities |
| 1               | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 |              |
| 0               | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 0 |              |
| 0               | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 |              |
| 0               | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 |              |
| Column parities |   |   |   |   |   |   |   |   |              |

a. Design of row and column parities

## 8.2.2 Cyclic Codes

### 10-4 CYCLIC CODES

***Cyclic codes** are special linear block codes with one extra property. In a cyclic code, if a codeword is cyclically shifted (rotated), the result is another codeword.*

#### [ Most Probable Question ]

. What are cyclic codes?

*Subclasses of linear block codes having the properties of cyclic shift of one code word to produce another code word i.e., shifting cyclically to left from MSB position to LSB position*

*Benefits: -simpler and easy to implement*

*-no requirement of additional memory for lookup storage*

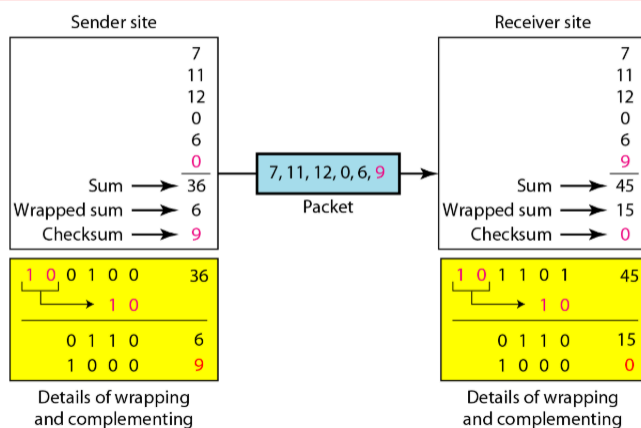
*-but error correction is complicated*

## 8.2.3 Checksum

## 10-5 CHECKSUM

*The last error detection method we discuss here is called the checksum. The checksum is used in the Internet by several protocols although not at the data link layer. However, we briefly discuss it here to complete our discussion on error checking*

Figure 10.24 Example 10.22

**Sender site:**

1. The message is divided into 16-bit words.
2. The value of the checksum word is set to 0.
3. All words including the checksum are added using one's complement addition.
4. The sum is complemented and becomes the checksum.
5. The checksum is sent with the data.

**Receiver site:**

1. The message (including checksum) is divided into 16-bit words.
2. All words are added using one's complement addition.
3. The sum is complemented and becomes the new checksum.
4. If the value of checksum is 0, the message is accepted; otherwise, it is rejected.

## 8.2.4 CRC-n ( $n = 3$ to $32$ )

CRC = Cyclic Redundancy Code

N = Degree of generator polynomial

Q. Explain How CRC Works ? for  $n = 3, 4, 5, 6, 7, 8$  . [ Most Probable Question ]

Figure 10.14 CRC encoder and decoder

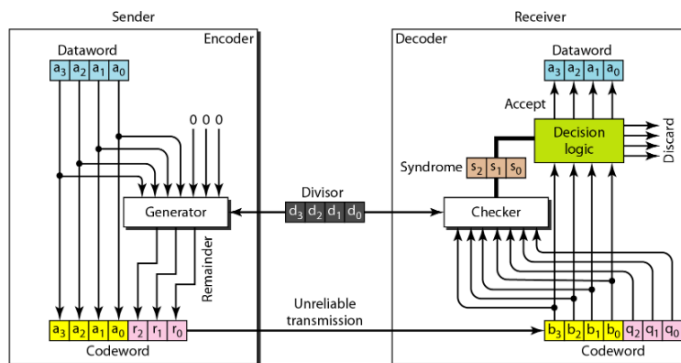
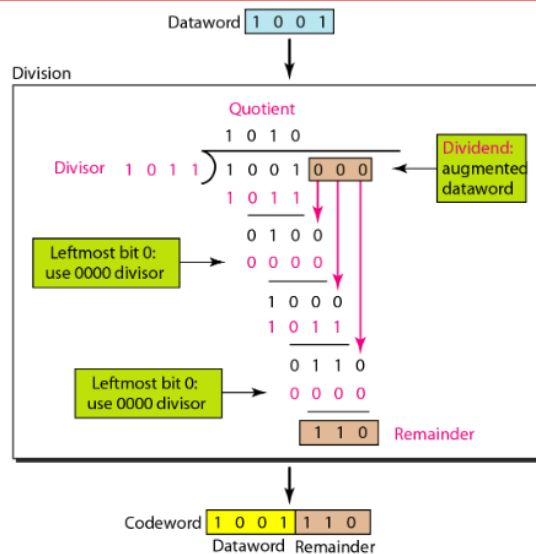
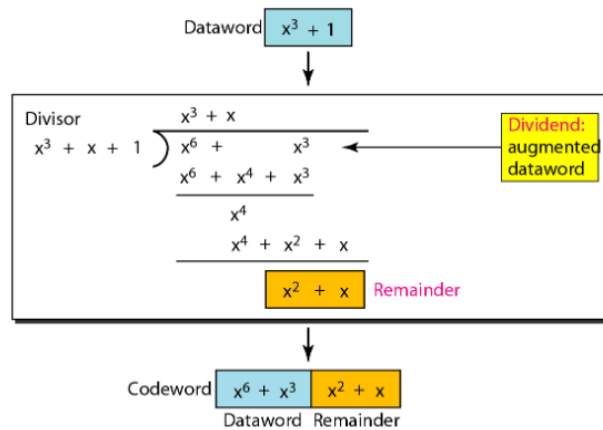


Figure 10.15 Division in CRC encoder



**Figure 10.22** CRC division using polynomials

The divisor in a cyclic code is normally called the generator polynomial or simply the generator.

If the generator has more than one term and the coefficient of  $x^0$  is 1, all single errors can be caught.

**Table 10.7** Standard polynomials

| Name   | Polynomial                                                                                                    | Application |
|--------|---------------------------------------------------------------------------------------------------------------|-------------|
| CRC-8  | $x^8 + x^2 + x + 1$                                                                                           | ATM header  |
| CRC-10 | $x^{10} + x^9 + x^5 + x^4 + x^2 + 1$                                                                          | ATM AAL     |
| CRC-16 | $x^{16} + x^{12} + x^5 + 1$                                                                                   | HDLC        |
| CRC-32 | $x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$ | LANs        |